

Koral Kulacoglu

Toronto, Canada | Canadian Citizen

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EDUCATION

University of Waterloo | Bachelor of Software Engineering (AI Specialization) 3.8/4.0 GPA | Sep 2023 – Apr 2028

- **Awards:** [USACO Platinum](#) [🔗](#), [Hack the North Winner](#) [🔗](#), [1st at Canada-Wide Science Fair](#) [🔗](#) & [2nd at EUCYS](#) [🔗](#)
- **Courses:** Advanced Algorithms, Operating Systems, Databases, Compilers, Signals & Systems, Numerical Computation

EXPERIENCE

Coinbase | Software Engineer Intern – [Institutional Financing](#) [🔗](#) San Francisco, CA | May 2026 – Aug 2026

- Built a **Go pipeline** (**Temporal**, **Kafka**) to compute **risk parameters** for **95+** assets and feed them to the **cross-margin** engine, cutting staleness from **24h to 30min** and removing **30%** average volatility drift (**86% p95** from margin sizing).
- Developed a **hot/cold storage model** for market data: **PostgreSQL** for **sub-10ms reads** on a rolling 90-day window, with daily prune and **Snowflake** ETL backup jobs for history retention.
- Added **Datadog latency tracking** to margin endpoints to flag margin fan-out as a bottleneck (**781ms p95**, $4.7\times$ the rest).

Optiver | FutureFocus Tech Chicago, IL | Mar 2026

- Engineered a deterministic **TCP-based electronic exchange** in **Python**, utilizing socket programming and unit testing.

BitGo | Software Engineer Intern – DevEx Palo Alto, CA | Sep 2025 – Dec 2025

- Built a **Rust static analysis** engine using **Tree-sitter** and **GraphQL** to **lazy-parse** package manifests, protecting **320+** BitGo repositories from supply chain attacks; parallelized via **shared-state concurrency**, cutting runtime by **95%** (5m to 15s).

AtoB | Software Engineer Intern – Growth San Francisco, CA | Jan 2025 – Apr 2025

- Created data pipelines for lead scoring, including a **Ruby on Rails UTM tracking system** ingesting **500K+** user interaction events monthly via **Sidekiq**, and a **Python Selenium** UCC-1 web scraper, collecting **\$150K+** worth of data into **Snowflake**.
- Utilized **Sentence Transformers** to automate PDF-to-SQL mapping for [Fuel Statement Analyzer](#) [🔗](#), saving **\$50/day**.

D2L | Software Engineer Intern – D2L Labs Toronto, ON | May 2024 – Aug 2024

- Built a **multi-agent framework** with a **FAISS vector DB** to simulate student workflows (e.g., assignments, discussion posts) for automated course QA, deployed on **AWS Step Functions**, **Lambda**, and **DynamoDB**; adopted as [BrightspaceBot](#) [🔗](#).

PROJECTS

🔗 **FIX Exchange** Apr 2026 – Ongoing

- Engineered a **C++ equity exchange** with **FIX 4.2** over **TCP**, price-time priority matching, and **UDP multicast** market data.
- Benchmarked on **AWS EC2 c6i.metal** with **28μs engine latency** (p99) and **27K orders/sec**; automated via **GitHub Actions**.
- Reduced p99 match latency **99.8%** through cache-optimized **absl::btree_map/flat_hash_map** containers, while eliminating kernel jitter via **pure-spin loops** (`_mm_pause`), **SCHED_FIFO** scheduling, **CPU pinning**, and **SPSC** lock-free queues.
- Built a persistent **SQLite** database with **write-ahead logging** and **async batch-processing** for deterministic state recovery.

🔗 **Partcl/HRT Macro Placement Challenge** May 2026 – May 2026

- Built a GPU-optimized, pure **PyTorch/NumPy** macro placement engine using **FFT electrostatic global placement** with **Lin-Kernighan k-opt** and **hierarchical Late-Acceptance Hill Climbing**, beating the RePlAc baseline by **30%**.
- Engineered a **bit-exact incremental** reimplement of the proxy cost (within **1e-3** of the oracle) with **delta updates** per macro move, cutting evaluation **2000×** (4s to 2ms), enabling **millions** of Lin-Kernighan/LAHC trial moves under **55min**.

SKILLS

- **Languages:** C/C++, Python, Rust, Go, Ruby, Java, JavaScript, TypeScript, SQL, Shell Script, Assembly, LaTeX
- **Tools:** Linux, Git, Docker, AWS, Kubernetes, CMake, GraphQL, Temporal, TensorFlow, PyTorch, NumPy, Rails, NodeJS